

SIR SYED UNIVERSITY OF ENGINEERING & TECHNOLOGY
COMPUTER SCIENCE & INFORMATION TECHNOLOGY DEPARTMENT

Spring 2026
Machine Learning (CS-466)
Assignment # 2

Semester: 8th
Announced Date: 14/05/2026
Total Marks: 03
Instructor: Ms. Tahmina Khan

Batch: 2022F
Due Date: 29/05/2026
Obtained marks: _____

Submitted by: _____ Roll #: _____ Section: _____

CLO #	Course Learning Outcomes (CLOs)	PLO Mapping	Bloom’s Taxonomy
2	To be able to apply different machine learning techniques to identify problems and provide their solutions.	PLO_3 (Problem Analysis)	C3 (Applying)

Question No. 01: [3 Marks]

Apply YOLO to the image containing multiple objects. Use the following settings:

- Anchor points: 5
- Anchor boxes: 3
- NMS threshold: 0.5
- Confidence threshold: 0.8

Image: [Insert any image with multiple objects]

Task:

- Preprocess the image and prepare it for YOLO input.
- Apply YOLO to detect objects in the image.
- Implement NMS to filter out duplicate detections.
- Visualize the detection results, including bounding boxes and class labels.
- Calculate the mAP (mean Average Precision) for the detection results.

Requirements:

- Use a latest YOLO variant and implement it using a deep learning framework (e.g., PyTorch, TensorFlow).
- Code should be written in Python.
- Provide a detailed report explaining the steps taken, results, and discussions.

Grading Criteria:

- Accuracy of object detection.
- Effective use of NMS and anchor points.
- Code quality and organization.
- Clarity of report and results